

Sales Intelligence & Business Analytics

Identifying Trends, Risks, and Opportunities

Sales Intelligence & Business Analytics

- DATA-DRIVEN INSIGHTS USING SQL & PYTHON , POWER BI
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 - Role: Data Analyst
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Project Overview

- This project focuses on analyzing sales data to identify key trends, patterns, and business insights. The objective is to understand customer behavior, product performance, and overall sales growth to support data-driven decision-making.
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Database & Tables Used

- Customers
 - Orders
 - Products
 - Transactions
 - Subscription Plan
 - Subscription Details
 - Ratings
 - Returns & Refunds
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Tools & Technologies

- ❑ **SQL** – Data Cleaning, EDA, Joins & Aggregations
 - ❑ **Python** – EDA & Visualization
 - ❑ **Libraries** – Pandas, NumPy, Matplotlib
 - ❑ **Power BI** – Data Modelling & Dashboards
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Data Cleaning & Preparation

- Removed duplicate records
 - Handled NULL & blank values –(string Values)
 - Validated relationships across tables
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Key Business Metrics

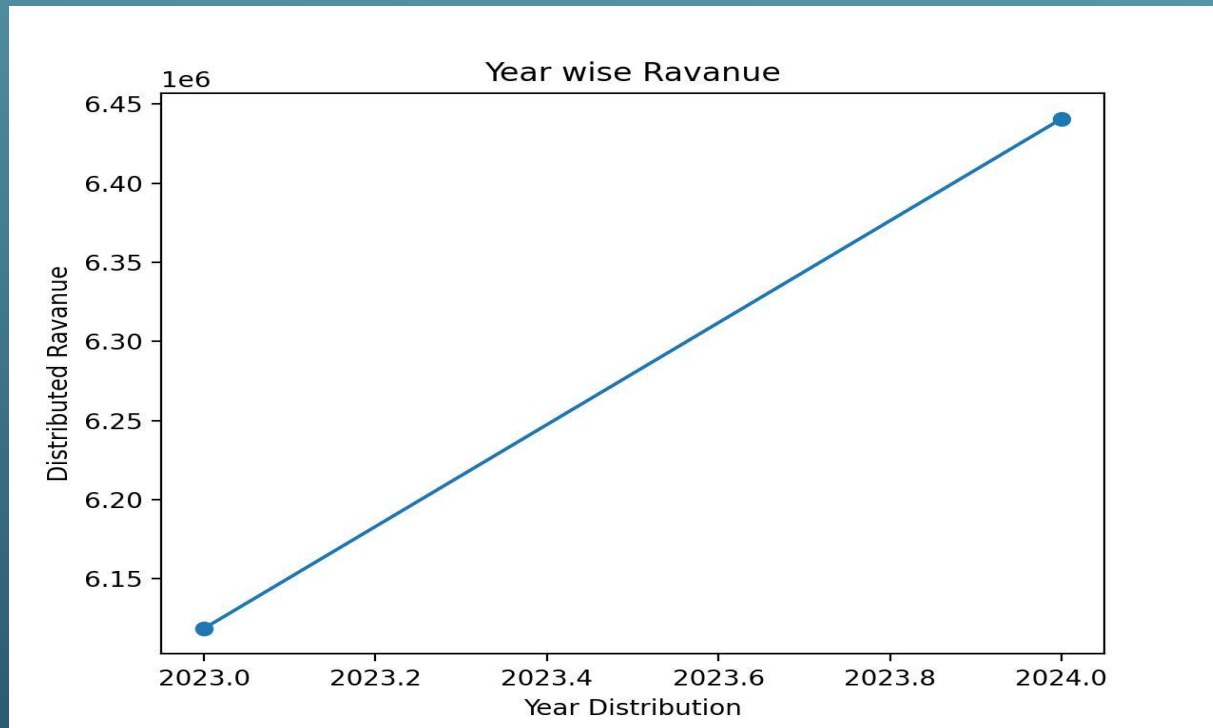
- Total Orders : 10,000
- Total Revenue : ₹ 11.46 Million
- Average Order Value (AOV) : ₹ 1229.36 in 2023
₹ 1282.23 in 2024
- Overall Return Rate: 48.45%

Sales & Revenue Analysis

➤ Analyzed revenue trends over time

Revenue grew from ₹6.12M (2023) to ₹6.44M (2024) → 5.26% increase.

show 2023 vs 2024 line chart & highlight 5.26% growth.

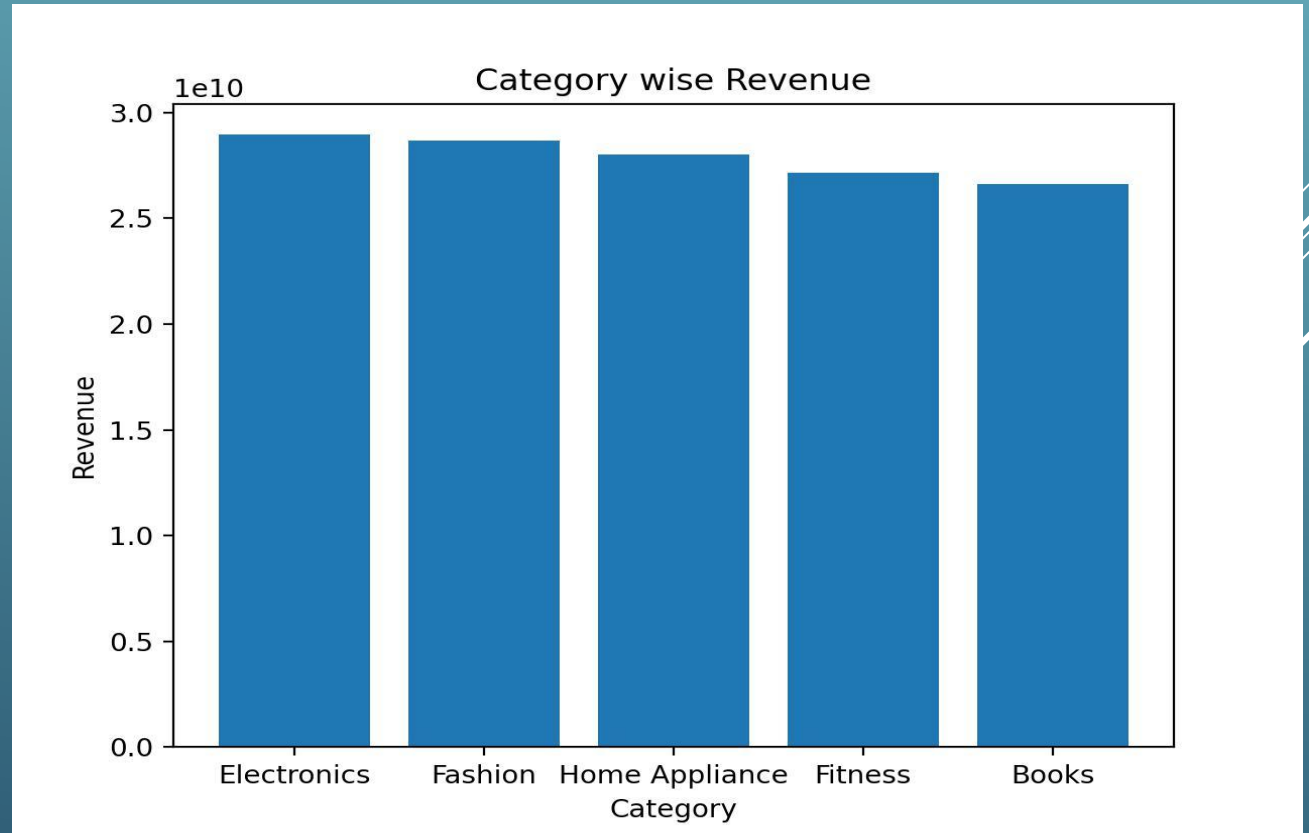


The 5.26% rise in revenue highlights effective sales execution and improved demand in 2024 compared to the previous year.

- Identified high-performing products and categories

This analysis found that balance performance across a category wise revenue with no standout high and low performer.

- Electronics category shows highest order volume
- Consistently strong revenue contribution



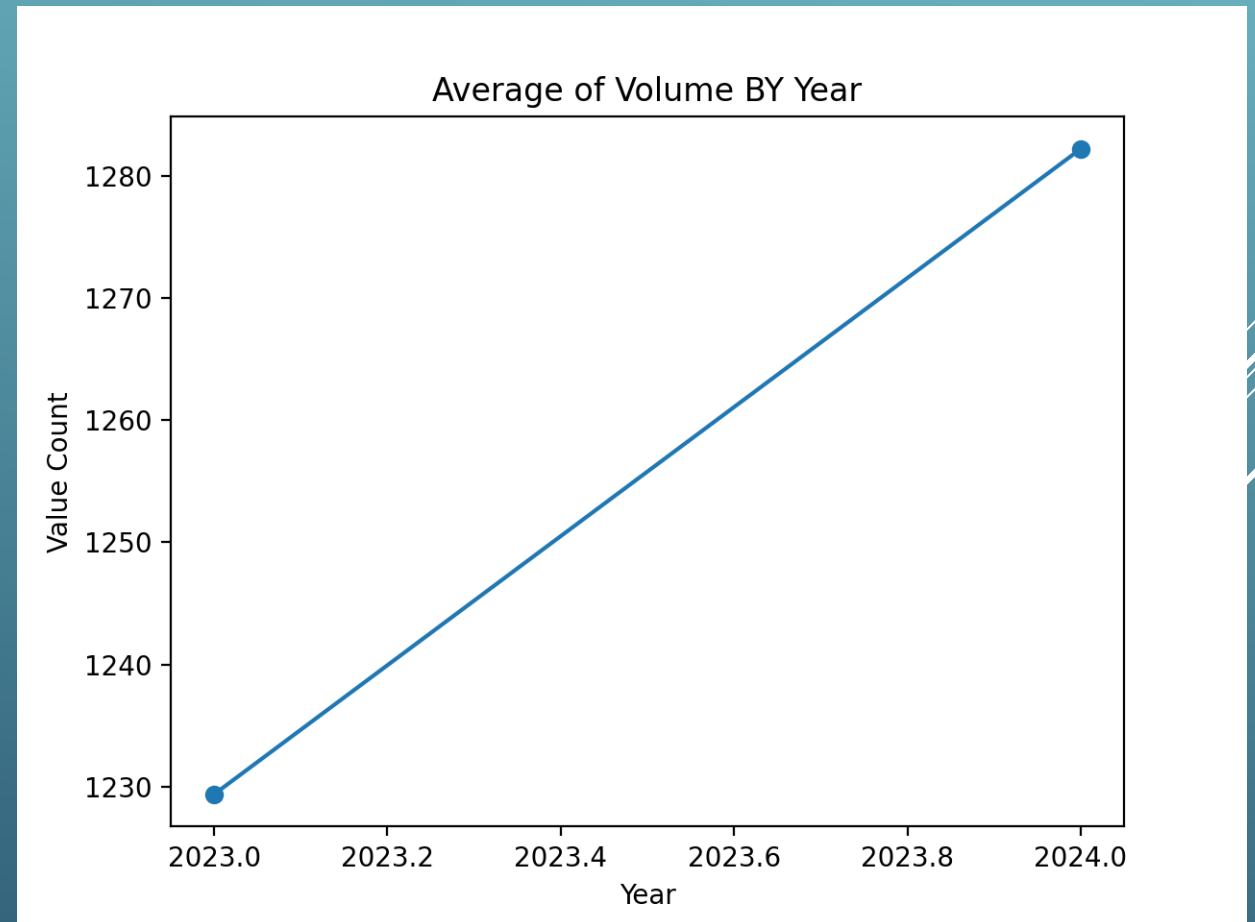
➤ Analyze Average Order Volume (AOV) VS Revenue

1. AOV (Average Order Value) is calculated:

- 2023 AOV = 1229.36
- 2024 AOV = 1282.23.2

2. Growth: AOV increased by ₹52.88 from 2023 to 2024.

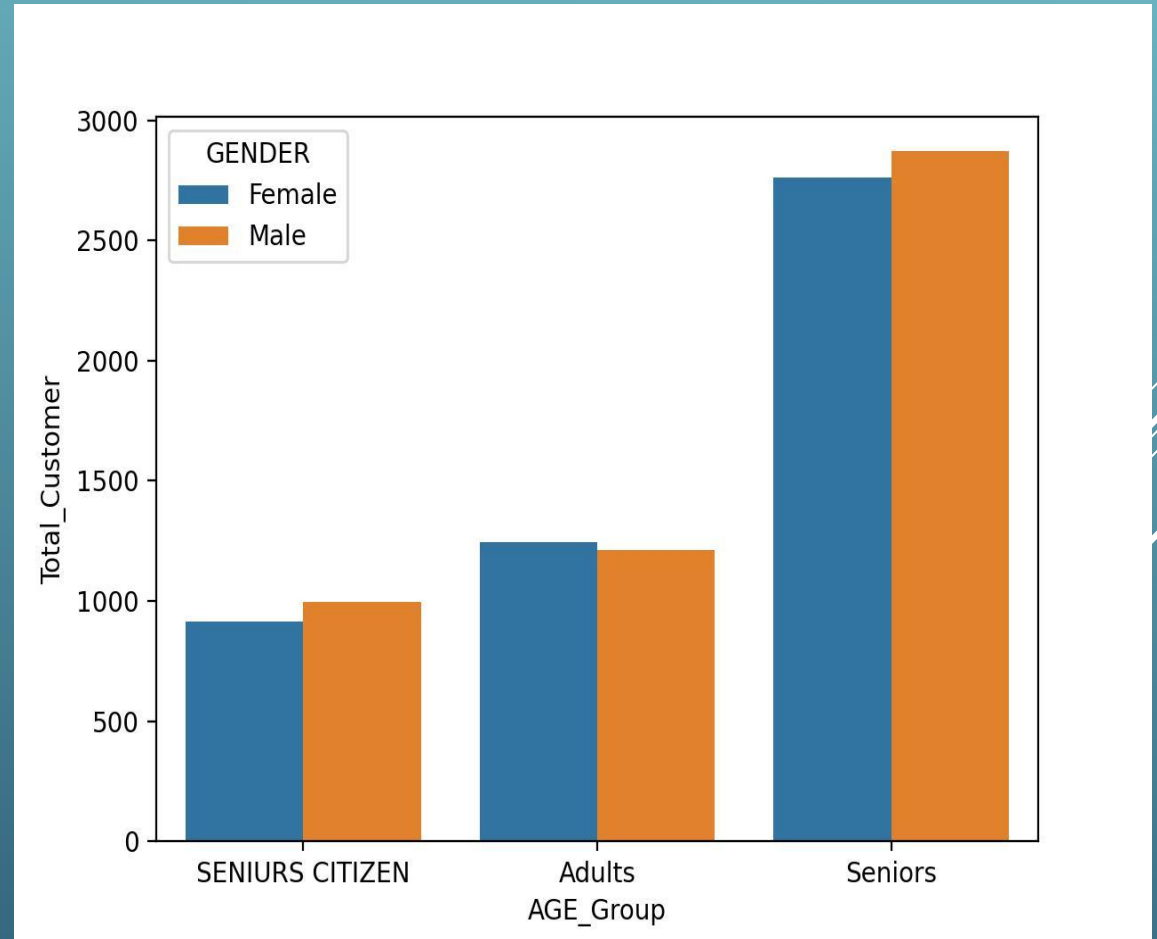
**The average order value Increased
Indicating customers are spending more
per order or order size is increasing.**



Customer Behavior Analysis

- The chart indicates age_group has three categories: SENIOR CITIZEN, Adults, and Seniors.
- Analyzed customer data by age, gender.
- Adult customers (18–30) contribute maximum orders

Seniors drive the highest customer volume (especially males), while SENIORS CITIZEN have the smallest segment with slightly more males than females.

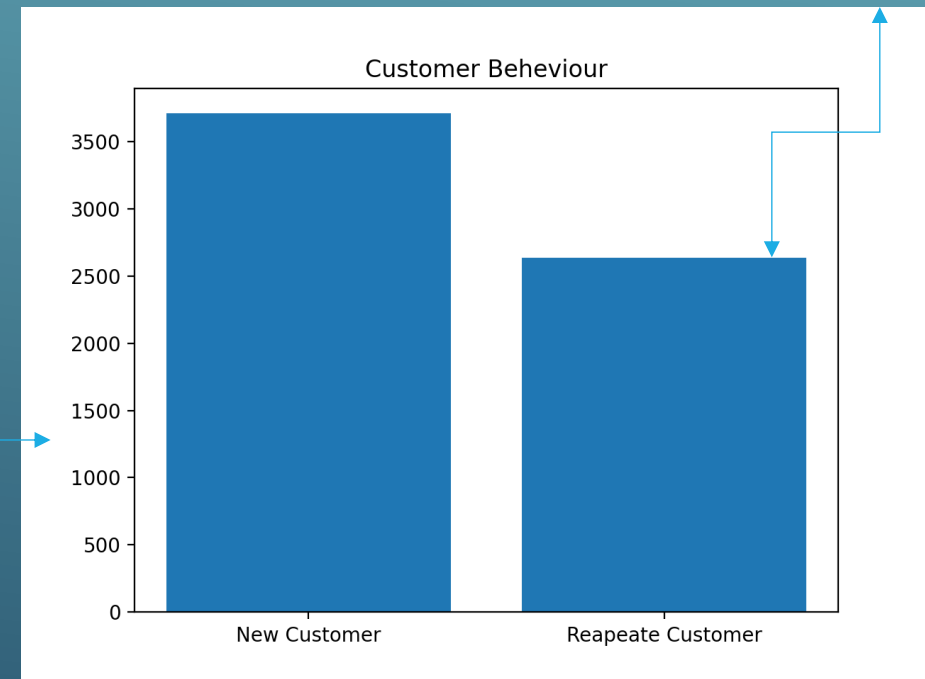


New vs Repeat Customers

- New Customers: 3,707
- Repeat Customers: 2,633
- Indicates good acquisition with retention improvement scope

New Customer
-- 3707

Repeat Customer
-- 2633



Subscription Analysis

➤ Analyzed subscription plans and usage

Plan Name	Feature	Total_Customer
Prime Premium	Free Delivery, 4K Streaming	3283
Prime Basic	Free Delivery, Limited Streaming	3360
Prime Standard	Free Delivery, Full HD Streaming	3393

Prime Standard is the most popular subscription plan, (there are 3,393 customers). followed closely by Prime Basic and Prime Premium.

Payment & Transaction Analysis

- Evaluated payment modes and success rates
- Net Banking observed as most used payment mode – 2024

Net Banking	2024
Debit Card	2011
Credit Card	1980
UPI	1978
Wallet	1974
NOT Mention	33

Net Banking is used the most , while 33 customer are “Not Mentioned” in payment mode & Other payment modes show nearly equal usage.

Ratings & Customer Satisfaction

- Compared ratings across categories
- Identified products with high sales but low ratings

Category	Quantity	AVG_Rating
Electronics	4059116.19	3.01487524
Fashion	3970766.2	2.967883212
Home Appliance	3707907.93	2.989180835
Books	3642881.96	2.982226869
Fitness	3620959.77	3.002556237

Customer satisfaction is highest in Electronics (avg rating 3.01) and lowest in Fashion (avg rating 2.96).

Returns & Refunds Analysis

➤ Overall Return Rate: 48.45%

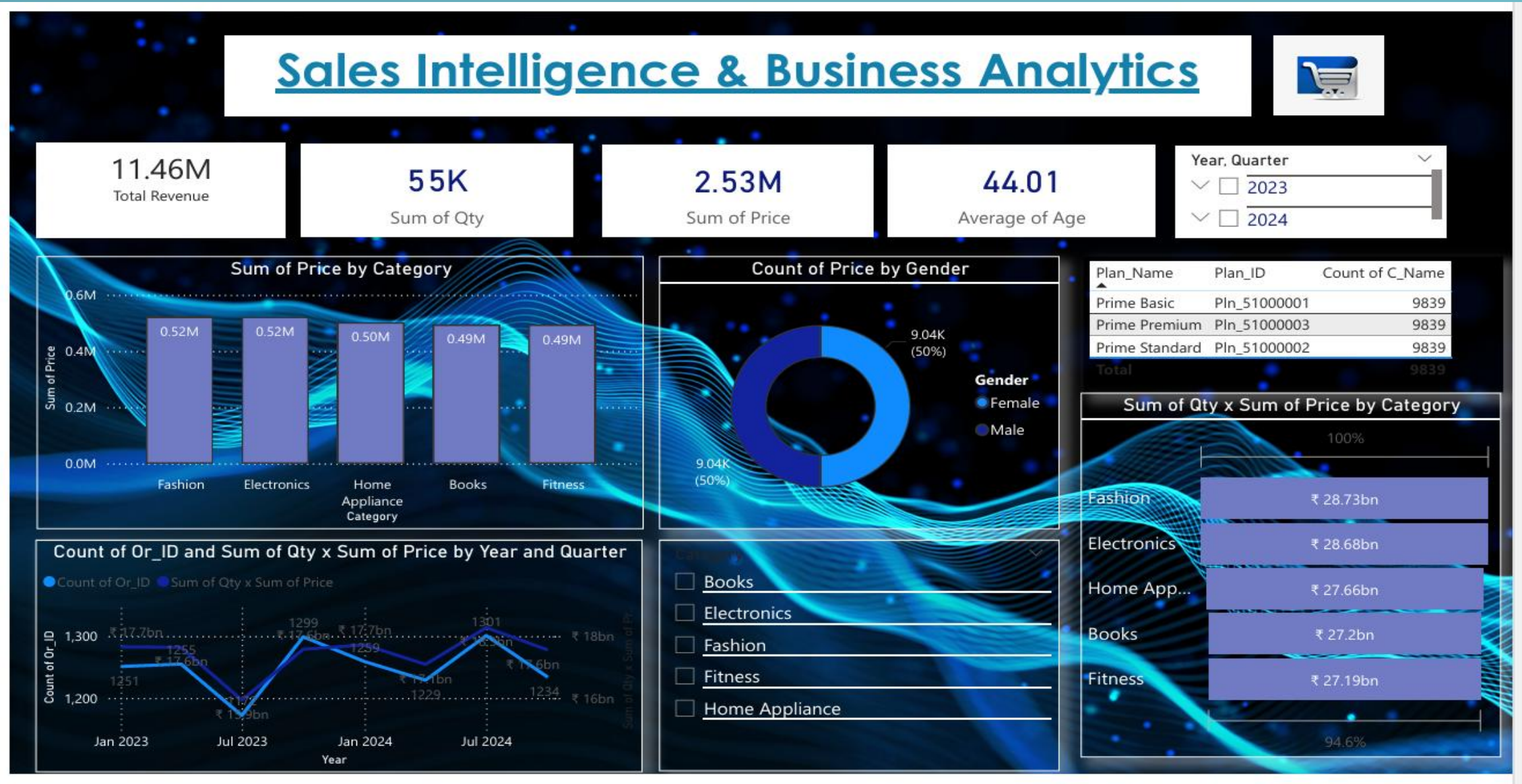
Category	Total Return
Home Appliance	52.83019
Fashion	51.08434
Electronics	47.59494
Fitness	47.1831
Books	43.7659

Home Appliances show
highest return
frequency

Business Recommendations

- Improve quality of low-rated products
 - Reduce returns via better descriptions and QC
 - Strengthen subscription-based retention
 - Monitor delivery and payment failures
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DashBoard Representation



Conclusion

- End-to-end sales data analysis completed using SQL ,Python & Dashboard.
 - Actionable insights generated across sales, customers, and post-purchase behavior
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THANK YOU

